

1. Explain the distinct roles of hemoglobin and myoglobin in the body and note the similarities of these two proteins. Include  $O_2$  transport vs  $O_2$  storage, tetramer vs. monomer, where each is located, etc.  
Hemoglobin is located in red blood cells (RBC) and carries  $O_2$  in the blood stream from the lungs to metabolically active tissue where the pH is lower. At lower pH, hemoglobin's binding to  $O_2$  is weaker, so  $O_2$  is released and can diffuse into nearby cells. Myoglobin is located at high concentration (about 100 mM) in the cytoplasm of cells and binds tightly to  $O_2$ , serving as a storage form of  $O_2$ . The solubility of free  $O_2$  in aqueous solution/blood is low (about 0.3 mM), so Mb significantly increases the amount of  $O_2$  stored in cells (by about 333-fold!). Hemoglobin is a tetramer (4 subunits), while myoglobin is a monomer (1 subunit). However, both hemoglobin and myoglobin have the same globin fold (an 8-helix, compact structure that binds heme using two His residues, a proximal His that binds covalently to the heme and a distal His that stabilizes bound  $O_2$ ).
2. Explain why we don't simply use free heme to store  $O_2$  in the body (i.e., why is myoglobin needed?).  
There are two reasons the myoglobin protein is needed. First, the free heme can form tetramers that react with  $O_2$ , oxidizing the heme iron into  $Fe^{3+}$  that binds to  $H_2O$ , not  $O_2$ . Second, the myoglobin effectively increases the concentration of stored  $O_2$  by more than two orders of magnitude, enabling the cell to have plenty of  $O_2$  available when needed.
3. Short answer questions:
  - a. What are the only three experimental methods that can determine protein 3D structure at atomic resolution?  
**X-ray crystallography, NMR, and cryo-EM**
  - b. Can the amino acid sequence of a protein enable prediction of that protein's 3D structure? Explain.  
**Yes. In general, AlphaFold, an artificial intelligence-based protein structure prediction tool, can now accurately predict the 3D structure of a protein from its amino acid sequence.**
  - c. Typically, are the structures of a protein determined by x-ray diffraction and by NMR in solution very different from each other? Why?  
**No, in most cases they are the same or very similar. This is because protein crystals are typically composed of ~50% water by volume, so the protein is in an environment that is similar to aqueous solution.**
  - d. Why did it take about 30 years to solve the first protein structure by x-ray diffraction?  
**The big challenge in the early days of x-ray crystallography was solving the phase problem, i.e., figuring out how to obtain the missing phases of the diffracted x-rays for which only the intensities can be recorded.**
4. Multiple Choice:
  - a. A photon (circle all that apply)
    - ☒ i. has no mass
    - ii. has mass equal to a proton mass (1 atomic mass unit, amu)
    - ☒ iii. has a distinct energy determined by its frequency
    - ☒ iv. behaves as both a wave and a particle
    - v. behaves only as a particle in the x-ray crystallography experiment
  - b. A protein primary structure (circle all that apply)
    - ☒ i. is the amino acid sequence
    - ☒ ii. can be used to construct a homology model
    - ☒ iii. can be used to study evolutionary relationships
    - ☒ iv. can be used to predict 3D structure by AlphaFold
    - v. is held together by hydrogen bonds
5. Circle T if true, F if false:
  - a. ☒ T / F The  $\phi, \psi$  values at the **i+1 and i+2** positions define the type of  $\beta$ -turn
  - b. T / ☒ F Gly is required at the **i+2** position of a **Type I** turn
  - c. ☒ T / F Proline is favored at the **i+1** position of **both Type I and Type II**  $\beta$ -turns
  - d. T / ☒ F collagen is an enzyme
  - e. ☒ T / F Fully mature collagen (loose ends cleaved, crosslinked) only occurs outside of cells
  - f. T / ☒ F SCOP uses state of the art microscopes to determine protein structure
  - g. T / ☒ F If we had only the Ramachandran plot for a protein, we would have all necessary info to calculate its tertiary (III) structure.
6. Circle the correct answer:
  - a) A topology diagram shows alpha-helix / beta-sheet connections
  - b) Scurvy is due to a deficiency of Vitamin E / niacin / Vitamin C which leads to the inability to hydroxylate Gly / Ala / Pro and weakens the structure of  $\alpha$ -helices / collagen /  $\beta$ -turns.
  - c) Peptide planes and side chains **alternate** orientations in  $\alpha$ -helices /  $\beta$ -strands.
  - d) Procollagen is formed inside cells / outside cells but has loose ends that are not in the  $\beta$ -sheet /  $\alpha$ -helix / triple helix.
  - e) The total number of unique protein folds is thought to be unlimited / limited.

- f) Mature collagen has crosslinks / phosphorylation catalyzed by lysyl oxidase and involves peptide bond formation / aldol condensation.
- g) A superfamily is a class of proteins with same fold / different fold / no known fold that vary in sequence identity from high to zero / all have high sequence identity / all have low sequence identity.

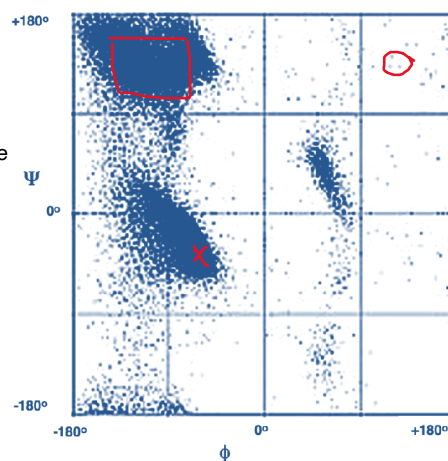
7. (2 pts) **Circle the letter** of the peptide that could be an amphipathic helix on the surface of a water-soluble protein. **Draw a square around the letter** of the peptide that could be inserted in a membrane. **Place an X next to the letter** of the peptide that could be found in collagen:

- ☒ a. Val-Val-Val-Val-Val-Val-Val-Val  
☒ b. Val-Asn-Val-Val-Asn-Val-Val-Val-Asn  
☐ c. Asn-Asn-Asn-Asn-Asn-Asn-Asn-Asn  
☒ d. Gly-Pro-Pro-Gly-Pro-Pro-Gly-Pro-Pro  
☐ e. Val-Asn-Asn-Asn-Asn-Asn-Val-Val

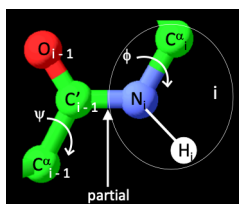
8. The Ramachandran Map on the right shows

- T / ☒ F the calculated Map for poly-Gly  
T / ☒ F regions that correspond to  $\alpha$ -helices and  $\beta$ -sheets, but not turns or collagen structure  
☒ T / F the observed values of  $\phi$  and  $\psi$  for real proteins, either hundreds of folded or unfolded

- Directly on this plot, draw an **X** where  $\phi, \psi$  for a **right-handed  $\alpha$ -helix** are located.
- Directly on this plot, draw a **circle** around an example of a region where only Gly can be located.
- Directly on this plot, draw **a box** around the region where  $\phi, \psi$  for a  $\beta$ -sheet are located.



9. Draw **ONLY THE SIX ATOMS** that are constrained to lie in a plane because of the properties of the peptide bond. Draw no other atoms! Label all six atoms. Show with an **arrow** and a **label** the bond about which the  $\phi$  rotation occurs, and the bond about which the  $\psi$  rotation occurs. **Show with an arrow labeled "PARTIAL"** the bond (other than a carbonyl) that has partial double bond character. **Circle the atoms in ONE amino acid residue**. If the atoms you circled are in amino acid residue "i", label the remaining three atoms (e.g., with either "i - 1" or "i + 1").



Alternatively, the left three atoms could have been circled and labeled with "i", in which case the right three atoms would be labeled "i + 1".

**Note** that in the peptide plane,  $\psi$  (rotation about  $C^\alpha-C'$  bond) corresponds to the amino acid on the N-terminal side of the plane, and  $\phi$  (rotation about  $N-C^\alpha$  bond) corresponds to the amino acid on the C-terminal side of the plane.

10. Match letter of each **image** with its correct description:

- ☒ B cis peptide bond  
☒ A trans peptide bond

**NOTE:** When might we expect to see a cis peptide bond? For the X-Pro peptide bond, in the trans isomer the  $\delta$  C occupies roughly the same position as the  $\alpha$ -C in the cis isomer. Thus, the free energy of the trans isomer is closer to the free energy of the cis isomer, and the difference in free energy between trans and cis states is much smaller relative to a non-Pro residue. The Boltzmann equation tells us that when this energy difference is smaller, the probability of being in the higher energy state will increase. So, the X-Pro peptide bond has significant probability of being in the cis state!

